AURA Robotics · Hinterland Hack 2026

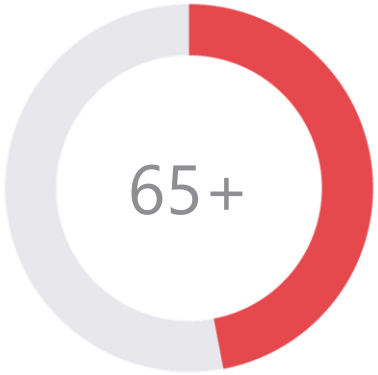
OttO

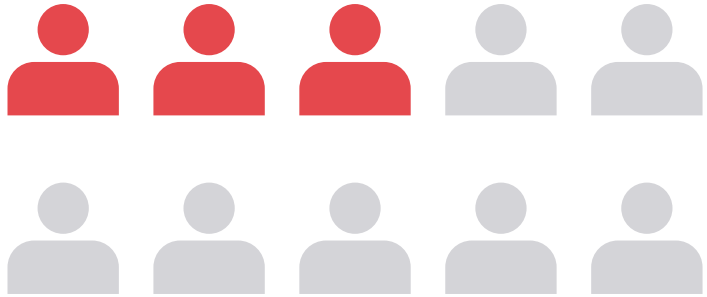
Your personal hospital guide.




Hospitals are where we bring the people we love.

Not just one story. It happens every day.







47 %

of hospital patients are over 65

29 %

of staff walk lost patients to the room

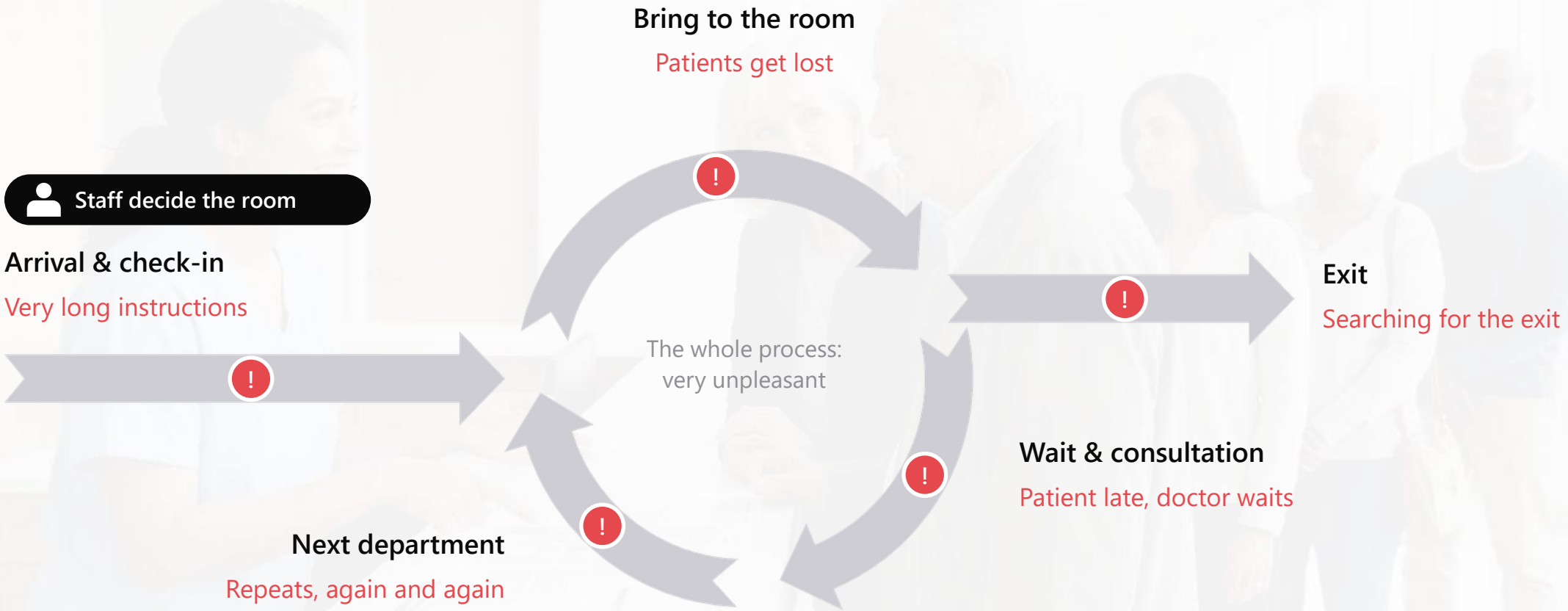
3 days

a year per staff member, not spent on care

Up to **690,000** nurses: the projected shortage in Germany by 2049.

Destatis 2024 (65+: own calculation) · Jamshidi et al., HERD 2025, 301 US hospital staff (3 days: 31.8 min/week × 46 weeks ÷ 8 h, own calculation) · Destatis nursing projection 2024: 280,000–690,000, all care settings

The patient journey today. Where it goes wrong.



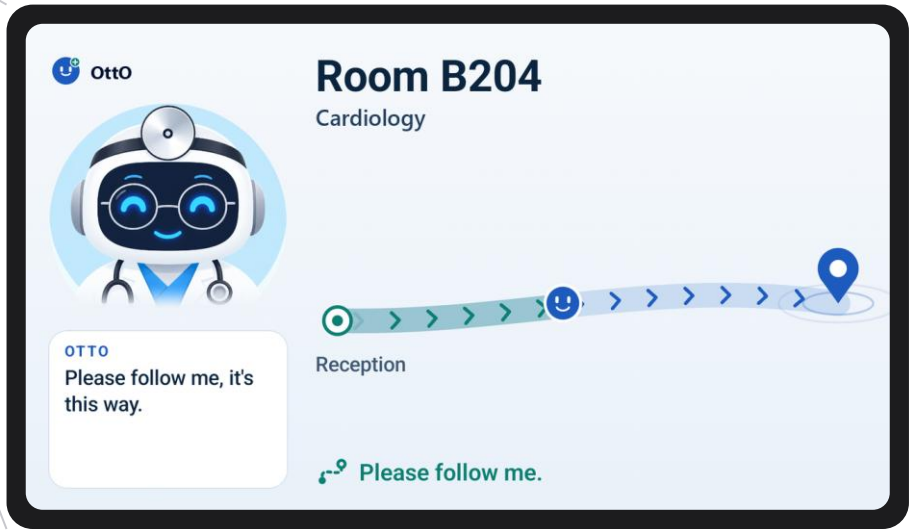
OttO.

Your hospital companion.

- Keeps an eye on you
- Adjusts to your pace
- No personal data



Walks with you



Reception decides where you go. OttO walks with you, like a buddy.

Same process, with OttO. From unpleasant to pleasant.

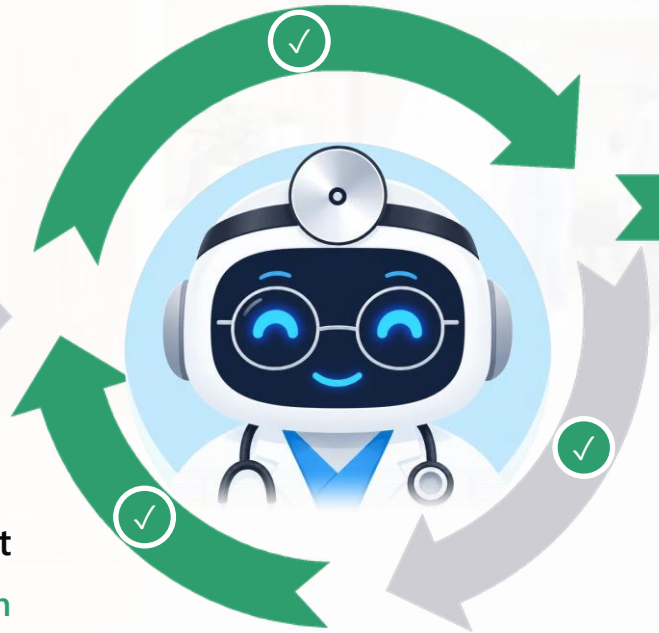
Staff decide the room

Arrival & check-in

No more route to explain

Bring to the room

OttO takes over



Exit

OttO walks you out

Wait & consultation

Doctor knows where you are

Next department

OttO picks you up again

People decide

OttO walks

OttO is for everyone.



Patients

Never get lost



Nurses & staff

Time back to care



Visitors

Guides your family to you

Running live on our robot today

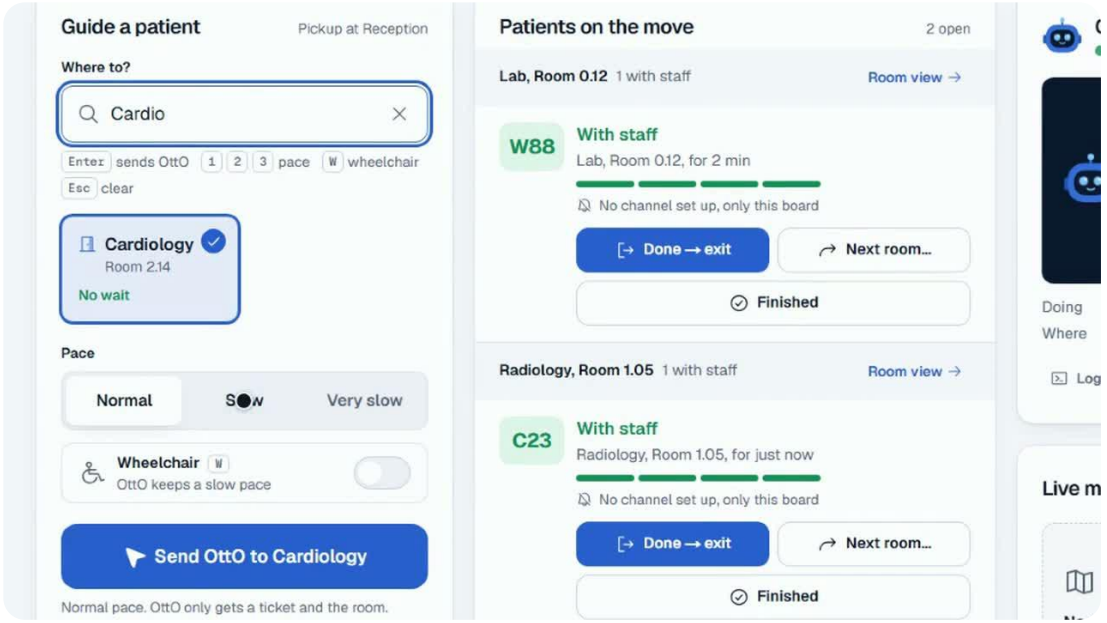
Live demo

From reception to the room, and back.



This is how it looks in the hospital.





From hackathon to hospital. 3, 6 and 12 months.







3 months

Pilot-ready

Technology ready

Regulations approved

6 months

First clinical pilot

With potential local partners:

Bethel · Franziskus Hospital

HDZ NRW · MHH

12 months

Scale

New regions

New use cases

New functions

Remember her from the film?

She isn’t lost anymore.

Let’s make sure your loved ones never walk a hospital alone.



This is us. Let's talk.



The Vinh Nguyen Trong
Construction robotics and AI, RWTH Aachen
LinkedIn →

OttO
Our robot guide

Sebastian Jack Maier
Cognitive computer science (AI), Bielefeld University
LinkedIn →

Dr. Evgenii Pustozerov
ML researcher, medical assistance systems
LinkedIn →

Backup. Situations OttO handles.

First visit, 78, with a walker

A nurse at reception sends her to cardiology. OttO walks at her pace and waits when she stops.

A visitor with flowers

He gives the patient's name to a person at reception, who looks up the room. OttO walks him there and never learns the name.

Monthly check-up, the room moved

Reception picks the new room. OttO knows the way, so the patient does not have to.

From consultation to X-ray

The doctor marks the visit as done. OttO picks the patient up, walks them to X-ray, then to the exit.

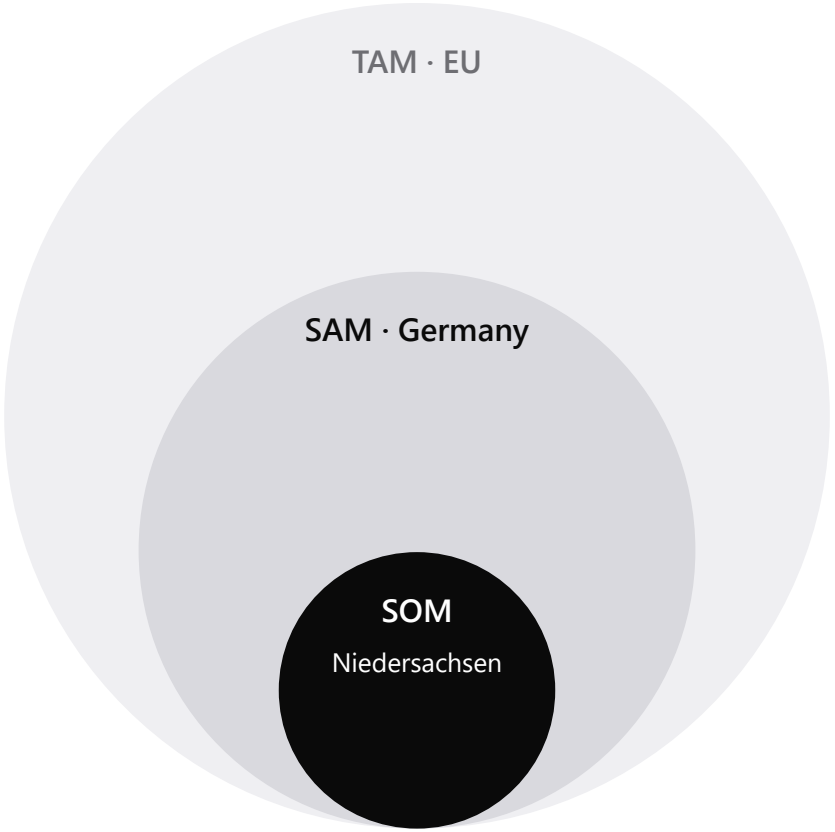
A wrong turn

OttO notices, waits, asks if everything is all right, and gets them back on the way.

People decide. OttO walks.

Reception and nurses greet you, take your data and decide where you go. OttO only knows the next room: no personal or clinical data.

Backup. Market size, bottom-up.



SOM ≈ €4M a year

Niedersachsen, our start: 165 hospitals × 2 robots × €12k

Niedersachsen state parliament, Drs. 19/4737 (hospitals end of 2023) · own calculation

SAM ≈ €44M a year

Germany: 1,841 hospitals × 2 robots × €12k

Destatis 2024 · own calculation

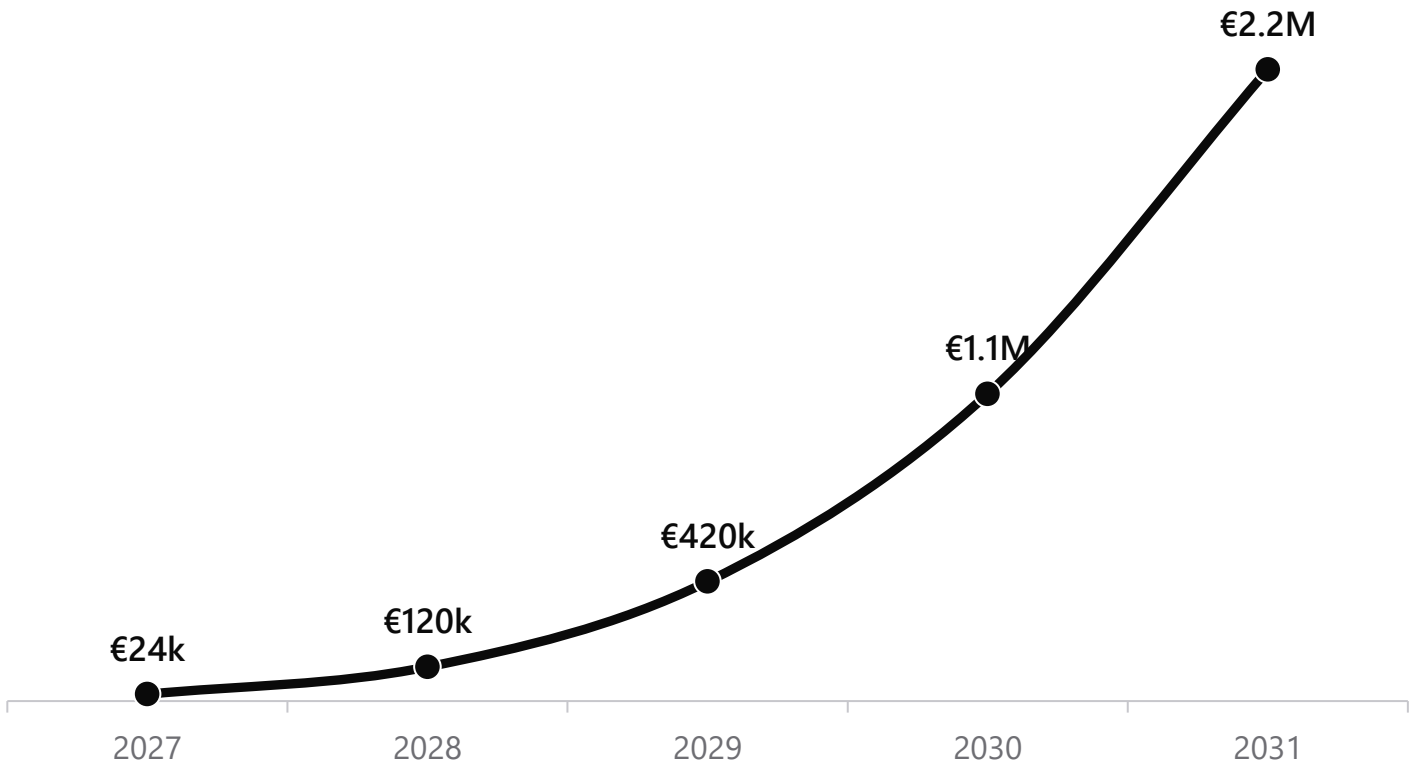
TAM ≈ €158M a year

EU: about 6,600 hospitals (Germany scaled by beds) × 2 robots × €12k

Eurostat 2022: 2.3M hospital beds in the EU, 642k in Germany · own calculation

Assumptions: 2 robots per hospital on average, €12k per robot per year incl. hardware, software and support. **Top-down check:** hospital wayfinding robots are about $1B worldwide in 2025 (The Business Research Company). Tailwind: medical robot sales +91 % in 2024 (IFR).

Backup. Growth plan, illustrative.



2027	**Pilot in OWL and Hannover** 2 robots
2028	**First paying clinics** 10 robots
2029	**Niedersachsen and NRW** 35 robots
2030	**Germany** 90 robots
2031	**≈ 5 % of German hospitals** 185 robots

Revenue at €12k per robot per year. A plan, not a forecast.

Backup. Go-to-market: land, expand, scale.

	2027 · first pilots **Land**	→	2028–29 · the region **Expand**	→	2030+ · Germany, then the EU **Scale**
Where	OWL and Hannover		Niedersachsen and NRW		Germany, then the EU
Who buys	1–2 pilot hospitals. Potential: Bethel, Franziskus, HDZ NRW, MHH		Large hospitals: many outpatients, long walks		Hospital groups and university hospitals
How we win	Founder-led. Nursing directors are our champions		Pilot results as proof. Hospital associations, purchasing groups		Sales and service through a robot partner. Trade fairs: DMEA, MEDICA
Goal	2 robots. Proof: staff time saved, fewer lost patients		10 → 35 robots		90 → 185 robots

Ecosystem Robot partner: hardware and service · Hospital IT: Teams, queue systems · Data protection and works council: sign-off · Nurses: champions

Robot-as-a-Service About €1,000 per robot per month: robot, software and support included. No upfront cost for the hospital.

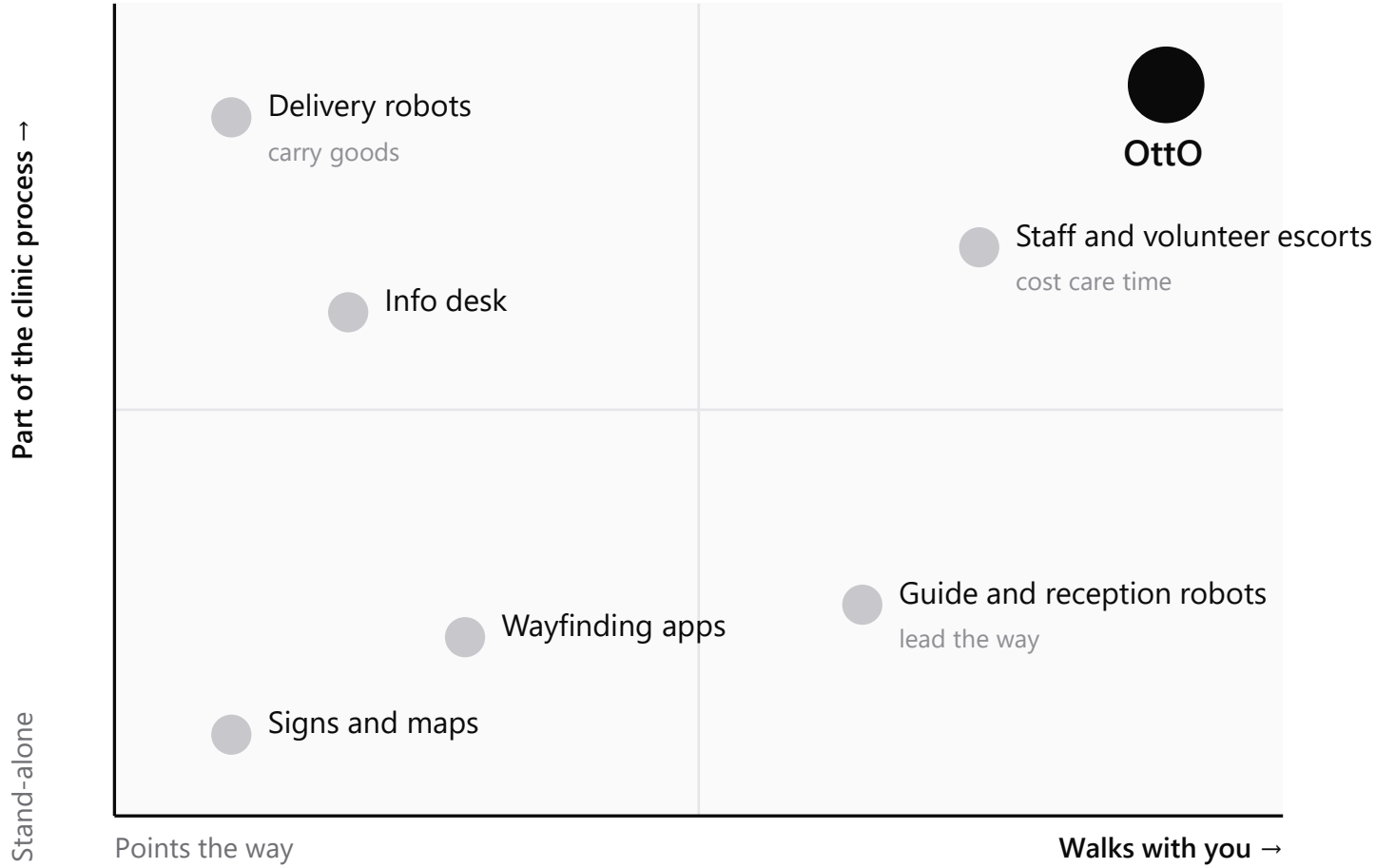
Backup. Competitor matrix.

	Walks with you	Adapts pace, notices if you fall behind	Part of the clinic process	No personal or clinical data	No staff time per guide
Signs, maps, info desk today's default	○	○	○	●	◐
Wayfinding apps e.g. Pointr, needs a smartphone	○	○	◐	◐	●
Staff and volunteer escorts costs care time	●	●	●	●	○
Guide and reception robots e.g. LG CLOi GuideBot, SoftBank Pepper	◐	◐	◐	◐	●
Delivery robots e.g. Aethon TUG, Keenon, Pudu: carry goods	○	○	◐	●	●
OttO AURA Robotics	●	●	●	●	●

● yes ◐ partly / depends on the product ○ no · Our view, based on public product information.

Our USP Of these options, only OttO walks with you like a person, fits into the clinic process, and needs no staff time and no personal data.

Backup. Where OttO fits.



Our USP

Walks with you like a person.

No staff time. No personal data.

Like an escort

walks at your pace and notices if you fall behind

Unlike an escort

no nurse has to leave the ward

Unlike other robots

part of the clinic process, from check-in to the exit

Built on

an existing service robot: we build the guide, not the hardware

Backup. The path to approval, step by step.

Step	What we do	With whom	When
Intended purpose	Write it down: OttO guides people. No diagnosis, no monitoring, no treatment. MDR qualification check (Art. 2(1), MDCG 2019-11): then likely not a medical device.	Regulatory advisor	**Month 1**
Data protection	Impact assessment (GDPR Art. 35), processing agreement (Art. 28), camera notice signs (Art. 13).	Hospital data protection officer	**Month 1–2**
Robot safety	Check the robot's CE marking; our software keeps its speed and stop limits. Risk assessment (ISO 12100, ISO 13482). EU Machinery Regulation applies from 20 Jan 2027.	Robot maker, safety expert	**Month 2**
AI Act	OttO talks with people: it must say clearly that it is an AI (Art. 50, applies since 2 Aug 2026).	AURA Robotics	**Month 2**
Hospital sign-off	IT security, hygiene and cleaning plan, fire and escape routes, works council.	Hospital IT, hygiene, works council	**Month 2–3**
Pilot contract	Pilot agreement, liability insurance, staff training, information for patients.	Hospital management	**Month 3**

For a guide robot there is no single "approval": it is these checks, most of them with the hospital. To be confirmed by a regulatory expert before the pilot.

Backup. Five challenges, five answers.

	Challenge		How we solve it
	**Approval** Medical device? Safety? AI Act?	→	✓ Guide only, no diagnosis. 3-month expert check.
	**Privacy** Cameras in a hospital worry people.	→	✓ No names, no clinical data, no video stored.
	**Busy corridors** Beds and crowds: robots get stuck.	→	✓ Slow, gives way, takes a detour, calls staff.
	**Acceptance** Will older patients trust a robot?	→	✓ People decide. OttO only walks.
	Budgets Tight budgets, slow purchasing.	→	✓ Pilot first, then a monthly fee. No upfront cost.

Backup. Numbers and sources.

Jamshidi et al., HERD 18(2), 2025, doi:10.1177/19375867251317240	301 US hospital staff, self-reported: mean 31.8 min/week on wayfinding help (median 15), 6.9 min per help, 29 % walk people all the way, ~44 % faced incivility. 3 days a year: 31.8 min × 46 weeks ÷ 8 h (own calculation).
Destatis, press releases 398 (Nov 2025) and 033 (Jan 2024)	1,841 hospitals, about 17.5 million inpatient cases in Germany, 2024. 280,000 to 690,000 nursing staff expected to be missing by 2049.
Destatis table 23131-b01 (own calculation)	47 % of 2024 inpatient cases were people aged 65+.
EDPB Guidelines 3/2019 · MDR (EU) 2017/745	Detecting a person without identifying them is not Art. 9 biometric data. Navigation-only is likely not a medical device (Art. 2(1), Recital 19); needs a proper check.
IFR World Robotics 2025, press release 7 Oct 2025	Almost 200,000 professional service robots sold in 2024 (+9 %); medical robots +91 % to about 16,700; robot-as-a-service fleet +31 %.
The Business Research Company, Smart Hospital Wayfinding Robot Global Market Report	$1.0B in 2025, $1.17B in 2026, $2.17B in 2030 (CAGR about 17 %). A single market-report estimate.
Illustrative value, not measured	25 walks a day × 3 min saved × 250 days ≈ 312 h staff time per robot per year (≈ €9,375 at €30/h). Robot cost not included.